The response from professional scientists was also rapid. Niels Bohr wrote a quick reply to the EPR paper, which many physicists claimed resolved all the puzzles. What is less clear is precisely how Bohr's paper was supposed to have achieved that; as brilliant and creative as he was as a thinker, Bohr was never an especially clear communicator, as he himself admitted. His paper was full of sentences like "in this stage there arises the essential problem of an influence on the precise conditions which define the possible types of prediction which regard the subsequent behavior of the system." Roughly, his argument was that we shouldn't go about attributing elements of reality to systems without taking into account how they are going to be observed. What is real, Bohr seems to suggest, depends not only on what we measure, but on how we choose to measure it.

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Einstein and his collaborators laid out what they took to be reasonable criteria for a physical theory—locality, and associating elements of reality to deterministically predictable quantities—and showed that quantum mechanics was incompatible with them. But they didn't conclude that quantum mechanics was wrong, just that it was incomplete. The hope remained alive that we would someday find a better theory that both was local and respected reality.

That hope was definitively squashed by John Stewart Bell, a physicist from Northern Ireland who worked at the CERN laboratory in Geneva, Switzerland. He became interested in the foundations of quantum mechanics in the 1960s, at a point in physics history when it was considered thoroughly disreputable to spend time thinking about such things. Today Bell's theorem on entanglement is considered one of the most important results in physics.

The theorem asks us to once again consider Alice and Bob and their entangled qubits with aligned spins. (Such quantum states are now known as *Bell states*, although it was David Bohm who first conceptualized the EPR puzzle in these terms.) Imagine that Alice measures the vertical spin of her particle, and obtains the result that it is spin-up. We now know that if Bob measures the vertical spin of his particle, he will also obtain spin-up. Furthermore, by the ordinary rules of quantum mechanics we know that if